WorldCAP: Controller-Adaptive Planning via BEV Latent World Model

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Abstract. End-to-end autonomous driving has advanced considerably with the development of comprehensive world models. However, most existing models either implicitly assume perfect, controller-agnostic execution or are tied to the specific controllers used to collect training data, failing to account for cross-controller heterogeneity. This mismatch introduces systematic biases in latent state transitions, undermining trajectory evaluation and planning reliability when the planner is paired with diverse controllers. We propose WorldCAP, a controller-conditioned framework that explicitly incorporates execution dynamics into latent BEV representations. A lightweight Controller Style Encoder (CSE) extracts intrinsic controller characteristics from observed tracking behaviors, which are then integrated into latent state evolution via a Controller-Aware Latent Alignment (CALA) mechanism. WorldCAP generates style-aware trajectory predictions that align latent evolution with realistic execution dynamics, enabling more faithful trajectory evaluation and robust planning across heterogeneous controllers. Experiments on the NAVSIM benchmark demonstrate that our approach improves trajectory fidelity, planning consistency, and generalization to diverse controller behaviors.

Keywords: Autonomous Driving · World Model · Model-Based Planning · Controller-Conditioned Dynamics · Latent State Transition

1 Introduction

End-to-end autonomous [3, 21, 29, 35] driving has recently shifted from single-trajectory planning [15, 16, 19] toward multi-modal trajectory generation [4, 20, 24] to better capture the inherent uncertainty in complex driving scenarios. Considering the multiplicity of plausible future trajectories, effective evaluation becomes critical to ensure safe and reliable decision-making, particularly under dynamic traffic conditions, occlusions, or sensor noise. Model-free approaches often rely on rule-based heuristics [6, 8, 37] or the current state for trajectory scoring [4, 20, 24], which fundamentally limits their ability to anticipate downstream consequences, making them prone to suboptimal or unsafe decisions in uncertain environments.

Inspired by human drivers who constantly anticipate multiple possible futures before executing maneuvers, model-based approaches leverage learned world models to predict future states conditioned on candidate trajectories, which is

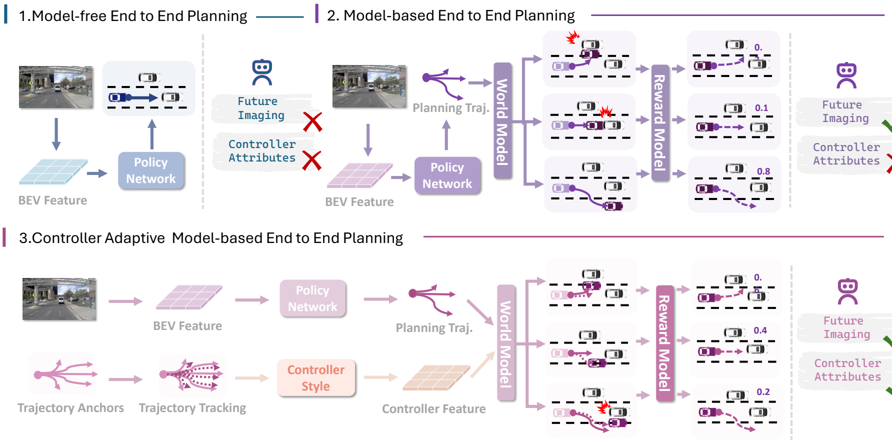

Fig. 1: Comparison of end-to-end planning paradigms. (1) Model-free end-to-end planning directly predicts a trajectory from the current BEV representation without explicit future reasoning. (2) Model-based end-to-end planning introduces a BEV world model to imagine future states for multiple candidate trajectories and performs reward-based selection, while implicitly assuming controller-agnostic execution. (3) Controller-adaptive model-based planning (WorldCAP) conditions latent BEV transitions on controller characteristics, producing controller-dependent future states that better reflect real execution dynamics and enable more reliable planning under heterogeneous controllers.

motivated by the success of world models in reinforcement learning [11,12,17,34], where they facilitate long-horizon planning and environment modeling [2,13,30]. Recent end-to-end autonomous driving methods incorporate world models to enhance trajectory evaluation by predicting future states with higher fidelity [10,14,22,31,38–40,44]. These models adopt various representations, for example: image-level, occupancy-grid, and bird's-eye view (BEV) latent spaces, each with distinct trade-offs in spatial abstraction and prediction efficiency. BEV representations are particularly attractive due to their low latency and spatial consistency, enabling single-step feed-forward future predictions that are more computationally efficient than the time-consuming multi-step denoising used in prior driving world models [14,38,39]. Moreover, BEV-based world models provide a unified representation compatible with downstream planning and control modules, facilitating more effective trajectory ranking and safer end-to-end decision-making.

Despite these advances, existing end-to-end driving systems that leverage world models typically learn latent dynamics tied to the specific controllers used during dataset collection. While these models implicitly capture the execution characteristics of the training controllers, they do not account for the heterogeneity of controllers or vehicles encountered at deployment. As a result, there is often a mismatch between the predicted future states and the actual vehicle execution, which can undermine trajectory evaluation and decision-making reliability.

In real autonomous driving systems, the realized future state is jointly determined by the planned trajectory, the tracking controller (e.g., gains, response characteristics, feedback policies), and vehicle post-dynamics (e.g., constant or scaling biases, actuator delays, low-pass filtered responses, and other structured lag effects). Neglecting controller heterogeneity introduces both stochastic and systematic deviations in latent state transitions, reducing the fidelity of imagined future states and affecting downstream planning decisions.

To address this challenge, we propose WorldCAP (Fig. 1), an end-to-end autonomous driving system that explicitly models execution-dependent dynamics. WorldCAP incorporates a Controller Style Encoder (CSE) to extract latent embeddings representing controller-specific execution characteristics from observed tracking behaviors. These embeddings capture systematic biases, response delays, gain mismatches, and other structured execution variations that arise from heterogeneous controllers or vehicle platforms. The extracted controller style vectors are then integrated with latent BEV representations through a Controller-Aware Latent Alignment (CALA) mechanism, which conditions the evolution of the latent world model on controller-specific characteristics at each prediction step. This alignment ensures that the predicted latent state trajectories reflect the cumulative effects of controller behavior. By explicitly conditioning the latent dynamics on controller style, WorldCAP generates predictions that are both consistent with the intended planner outputs and faithful to likely real-world execution. Furthermore, the modular design of WorldCAP allows it to generalize naturally to novel controllers or vehicle platforms without retraining the core planner, effectively bridging the gap between planned trajectories and actual vehicle execution while maintaining fidelity in latent state evolution and downstream reward modeling.

Experiments on the NAVSIM benchmark demonstrate that WorldCAP accurately captures controller-specific latent state evolution, thereby enhancing end-to-end planning performance. Our main contributions are summarized as follows:

- We highlight controller-dependent execution dynamics as a critical, previously overlooked factor in world-model-based trajectory evaluation, emphasizing their impact on latent BEV transition fidelity and planning–execution consistency.
- We present WorldCAP, a controller-conditioned end-to-end autonomous driving system that leverages a lightweight Controller Style Encoder (CSE) and Controller-Aware Latent Alignment (CALA) to produce more faithful, style-aware latent state predictions for trajectory evaluation and planning.
- Extensive NAVSIM experiments demonstrate that WorldCAP preserves high latent transition fidelity and robust trajectory ranking across diverse controllers, establishing controller-aware latent modeling as essential for reliable end-to-end autonomous driving.

2 Related Work

2.1 End-to-End Autonomous Driving

End-to-end autonomous driving [4,16,18,19,21,23,25,33,43,47] aims to learn a direct mapping from raw sensor inputs to future trajectories or control commands, circumventing explicit decomposition into perception, prediction, and planning modules. Existing approaches can be broadly categorized into imitation-learning-based and reinforcement-learning-based paradigms.

Imitation Learning. This remains the dominant paradigm, where predicted trajectories are supervised by expert demonstrations. Early methods leverage intermediate representations to facilitate planning, such as semantic occupancy grids or BEV features. For instance, P3 [32] adopts differentiable semantic occupancy as an intermediate scene representation, while ST-P3 [15] jointly learns spatial-temporal features across perception, prediction, and planning. UniAD [16] demonstrates the effectiveness of multi-task integration for robust goal-oriented planning, and VAD [19] introduces compact vectorized scene representations to reduce computation while preserving expressiveness. Subsequent works extend the single-trajectory paradigm toward multi-modal prediction to better capture inherent driving uncertainty. Diffusion-based generative policies, exemplified by DiffusionDrive [25], employ truncated diffusion with anchor-guided Gaussian priors and cascade decoders, enabling multi-modal trajectory prediction in real time. DiffusionDriveV2 [47] further integrates reinforcement learning to suppress low-quality modes and encourage exploration, balancing diversity and trajectory quality.

Reinforcement Learning. RL-based approaches [9,28,36,41,42,47] optimize policies using reward signals. RAD [9] leverages a 3D Gaussian Splatting (3DGS)-based closed-loop framework to reconstruct photorealistic environments for large-scale policy exploration. ReconDreamer-RL [28] incorporates video diffusion priors and dynamic adversary agents in reconstructed simulators, together with trajectory augmentation, to improve robustness under corner-case scenarios.

Limitations. Despite these advances, most end-to-end methods focus on improving perception-to-planning mapping or multi-modal trajectory generation. They generally assume ideal execution, neglecting the impact of tracking controllers or vehicle-specific post-dynamics. As a result, the discrepancy between planned and realized trajectories remains largely unaddressed, which limits performance in real-world deployment.

2.2 World Models for End-to-End Autonomous Driving

World models have been increasingly adopted in end-to-end autonomous driving to enhance scene understanding and facilitate trajectory planning. By learning structured representations of the environment, these models can replace or augment auxiliary perception and prediction tasks, enabling more efficient and robust planning.

For instance, OccWorld [45] employs a unified occupancy-centric representation to capture spatial-temporal scene dynamics, improving planning reliability. Drive-WM [39] generates high-fidelity driving videos through joint spatial-temporal modeling, providing rich supervision for planning modules. SSR [27] converts dense BEV features into sparse scene queries, leveraging world model representations to enhance scene understanding. LAW [21] adopts a latent world model framework and demonstrates effectiveness under both perception-free and perception-based settings. World4Drive [46] focuses on multi-modal trajectory generation and uses the world model to select the most suitable candidate, supporting robust trajectory evaluation under uncertainty.

Limitations. While these approaches excel at environment modeling and trajectory prediction, they either neglect execution-level controller dynamics entirely or learn representations specific to the training dataset controllers, limiting their ability to generalize to heterogeneous controllers and undermining execution-aware planning and trajectory evaluation.

3 Method

In this section, we present the core components of WorldCAP: the Trajectory Planner (Sec. 3.1), Controller Style Extractor (CSE, Sec. 3.2), Controller-Conditioned BEV World Model (Sec. 3.3), Reward Model (Sec. 3.4), and associated Training Losses (Sec. 3.5). As illustrated in Fig. 2, WorldCAP unifies multi-modal trajectory generation, controller-aware future state prediction, and reward-based trajectory evaluation. The Trajectory Planner generates candidate trajectories, while the CSE encodes intrinsic controller characteristics. These embeddings are incorporated into the BEV World Model through the Controller-Aware Latent Alignment (CALA) mechanism to ensure controller-conditioned state evolution. The Reward Model evaluates the predicted futures to score each candidate trajectory, and the highest-scoring trajectory is selected for execution, maintaining alignment between planning and downstream control dynamics.

3.1 Trajectory Planner

The Trajectory Planner generates multi-modal trajectory candidates conditioned on the current scene representation. Given multi-modal sensory observations, we first construct a unified BEV representation and then perform anchor-based residual refinement to obtain diverse trajectory proposals.

BEV State Encoding. Inspired by TransFuser [29], we leverage LiDAR and multi-view RGB images as inputs and encode them into a unified bird's-eye-view (BEV) representation. A BEV encoder projects the fused observations into a latent BEV feature map $B \in \mathbb{R}^{h \times w \times c}$, where h and w denote the spatial

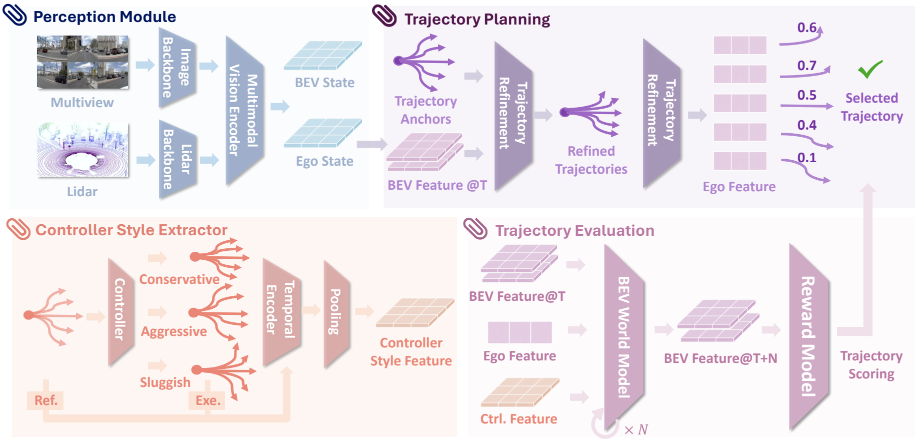

Fig. 2: Overview of the WorldCAP framework. WorldCAP is a controller-conditioned end-to-end autonomous driving system that integrates execution dynamics into world-model-based planning. The framework consists of four components: (1) a perception module that fuses multi-view camera and LiDAR inputs into unified BEV and ego-state representations; (2) a Controller Style Extractor (CSE) that encodes intrinsic controller characteristics from reference tracking trajectories; (3) a style-conditioned trajectory planner that generates and refines candidate anchors; and (4) a controller-aware BEV world model with a reward function for future prediction and trajectory ranking over a horizon of $T+N$. By conditioning latent state transitions on controller style, WorldCAP reduces execution mismatch bias and improves planning fidelity under diverse controller dynamics.

height and width of the BEV grid, and c represents the channel dimension. This latent BEV feature serves as the structured scene representation for downstream trajectory generation and future state prediction.

Multi-Modal Anchors. To capture diverse driving intentions, Following methods in [19, 24, 25], we adopt a set of trajectory anchors obtained via K-Means clustering over expert trajectories in the training set. These anchors represent representative motion modes in the data distribution and provide coarse initializations for multi-modal planning. We denote the anchor set as $\tau = \{\tau_i\}_{i=1}^{N}$.

Residual Refinement. Rather than predicting trajectories from scratch, we learn residual corrections over a set of predefined anchors conditioned on the BEV feature map, as shown in Fig. 2. Each anchor trajectory τ_i is independently encoded into a latent feature via a trajectory encoder $\mathrm{TE}(\cdot)$ implemented as an MLP. The encoded anchor feature serves as a query that attends to the BEV feature map B through cross-attention, enabling each trajectory proposal to adapt to the scene context. The attended feature is then passed through an MLP head to predict a trajectory-specific residual offset $\Delta\tau_i$, which is added to the corresponding anchor:

$$\hat{\tau}_i = \tau_i + \mathrm{MLP}\big(\mathrm{CrossAttention}(\mathrm{TE}(\tau_i), B, B)\big), \quad (1)$$

where $\hat{\tau}_i$ denotes the refined trajectory for the i-th anchor. The final trajectory set $\hat{\tau} = \{\hat{\tau}_i\}$ consists of all refined anchors.

3.2 Controller Style Extractor

In this section, we introduce a Controller Style Extractor (CSE) to encode intrinsic controller characteristics into a compact latent representation. In practical autonomous driving systems, the realized vehicle motion depends not only on the planned trajectory but also on the tracking controller and vehicle post-dynamics. Modeling such execution-level variation is therefore essential to ensure consistency between planning and control.

Residual Feature Construction We assume access to paired planned and executed trajectories collected from closed-loop simulations or real vehicles. Each trajectory is represented as a temporal sequence of ego states, where (x_t, y_t) denotes the vehicle position in the global frame and θ_t denotes the heading angle at time step t. Formally, a trajectory is denoted as:

$$\rho = \{(x_t, y_t, \theta_t)\}_{t=1}^{T}, \quad (2)$$

where ρ corresponds to either the reference trajectory ρ^{ref} or its executed counterpart ρ^{exec}.

To capture execution-level dynamics, we compute additional kinematic quantities from the raw state sequence, including linear velocity, acceleration, curvature, and heading rate via finite differences. These quantities, together with positional and angular deviations, are used to construct a per-timestep residual feature vector:

$$\Delta\rho_t = \phi(\rho_t^{exec}, \rho_t^{ref}), \quad (3)$$

where $\phi(\cdot)$ encodes lateral deviation in the Frenet frame, heading error, speed difference, curvature discrepancy, acceleration, and estimated steering rate. This structured representation captures both controller-dependent response characteristics (e.g., gain and damping behavior) and post-dynamics effects (e.g., constant bias, scaling bias, and temporal lag), forming the foundation for controller style analysis.

Controller Style Embedding The sequence of residual features $\{\Delta\rho_t\}_{t=1}^{T}$ is processed by a temporal encoder to produce a compact controller style embedding:

$$z_c = \mathrm{CSE}(\{\Delta\rho_t\}_{t=1}^{T}), \quad z_c \in \mathbb{R}^d. \quad (4)$$

Each per-timestep feature $\Delta\rho_t$ is projected into a hidden space via a small MLP and processed by 1D convolutions to capture local temporal patterns. To model long-range dependencies, a Transformer encoder with multi-head self-attention is applied, followed by temporal pooling and a final MLP projection to produce z_c. This embedding encodes intrinsic controller and post-dynamics characteristics and is used to condition the BEV world model for execution-aware trajectory prediction.

3.3 Controller-Conditioned BEV Latent World Model

Inspired by WoTE [22], Our controller-conditioned world model predicts future BEV states recurrently from state-action pairs while incorporating controller-dependent execution dynamics.

At each timestep, refined trajectories $\{\hat{\tau}^i\}$ are encoded into ego action embeddings $\{a_t^i\}$, which are combined with the latent BEV state B_t to form input pairs $(\tilde{B}_t, a_t^i)$. The BEV features capture static environment and vehicle context, whereas the action embeddings represent dynamic trajectory intent.

To model controller-dependent behavior, the latent BEV features B_t are conditioned on the extracted controller style embedding z_c via cross-attention:

$$\tilde{B}_t = \mathrm{CrossAttn}(B_t, z_c), \tag{5}$$

where B_t serves as the query and z_c provides key and value features. The conditioned BEV features $\tilde{B}_t$ are then combined with the corresponding action embeddings a_t^i and fed into a Transformer encoder to recurrently predict the next-step BEV states B_{t+1}^i and action embeddings a_{t+1}^i:

$$(B_{t+k+1}^i, a_{t+k+1}^i) = \mathrm{WorldModel}(\tilde{B}_{t+k}^i, a_{t+k}^i), \quad k = 0, \ldots, K-1. \tag{6}$$

By injecting controller style information at each timestep, the predicted BEV states encode execution-aware dynamics such as gain mismatch, actuation delay, and systematic control biases, while preserving the action embeddings that represent trajectory intent. This design enables accurate trajectory evaluation under heterogeneous controllers without modifying the core planner or reward model.

3.4 Reward Model and Trajectory Evaluation

WorldCAP predicts a reward for each candidate trajectory based on both the current BEV state and future latent BEV states generated by the controller-conditioned world model. Formally, let $B_t \in \mathbb{R}^{h\times w\times c}$ denote the current BEV state at time t, and $a_t^i \in \mathbb{R}^d$ the corresponding trajectory embedding produced by the trajectory predictor. The Reward Model predicts r_i using information from all future K steps, aggregating spatial and temporal context with 2D convolutions, followed by global pooling and an MLP head.

$$r_i = \text{RewardModel}(B^i_{\text{all}}, a^i_{\text{all}}), \tag{7}$$

$$B^i_{\text{all}} = \text{cat}\big([B_t, B^i_{t+1}, \ldots, B^i_{t+K}]\big), \tag{8}$$

$$a^i_{\text{all}} = \text{MLP}\Big(\text{cat}\big([a^i_t, a^i_{t+1}, \ldots, a^i_{t+K}]\big)\Big), \tag{9}$$

$$S_i = \text{cat}([B_{\text{pool}}, a^i_{\text{all}}]) \in \mathbb{R}^{2c}, \quad r_i \in \mathbb{R}^m. \tag{10}$$

Finally, the refined trajectory with the highest predicted reward is selected as the output trajectory.

3.5 Controller-Aware Future BEV Supervision

To enable execution-aware trajectory prediction, we condition future BEV supervision on controller-executed trajectories. For each planned trajectory, a target future BEV map M^{fut} is dynamically generated from a cached base map M^{base} according to the corresponding executed trajectory τ^{exec}:

$$M^{\text{fut}} = \mathcal{F}\big(M^{\text{base}}, \tau^{\text{exec}}\big). \tag{11}$$

The model predicts $\hat{M}^{\text{fut}}$, and supervision is applied via a focal loss [26]:

$$\mathcal{L}_{\text{fut-BEV}} = \text{FocalLoss}(\hat{M}^{\text{fut}}, M^{\text{fut}}). \tag{12}$$

This approach allows the model to learn scene occupancy conditioned on realistic, controller-executed trajectories, enabling execution-aware predictions without altering the core training pipeline.

4 Experiments

4.1 Benchmark

NAVSIM Dataset and Metrics. We evaluate our model on the NAVSIM dataset [7], a planning-centric benchmark built on OpenScene [5], a compact version of nuPlan [1]. NAVSIM provides high-fidelity driving scenes with synchronized sensor data from eight cameras covering 360° and a merged LiDAR point cloud from five LiDARs. The dataset is split into Navtrain (1,192 scenes) and Navtest (136 scenes). To assess model performance in realistic closed-loop driving, NAVSIM introduces the Predictive Driver Model Score (PDMS), which integrates No At-Fault Collision (NC), Drivable Area Compliance (DAC), Ego Progress (EP), Time-to-Collision (TTC), and Comfort (Comf.). The composite PDMS is computed as:

$$\text{PDMS} = \text{NC} \times \text{DAC} \times \frac{5 \cdot \text{EP} + 5 \cdot \text{TTC} + 2 \cdot \text{Comf.}}{12}, \tag{13}$$

providing a holistic measure of trajectory quality that balances safety, operational efficiency, and passenger comfort.

4.2 Implementation Details

We adopt a multi-view input, combining the front camera with center-cropped front-left and front-right images and a 64m × 64m LiDAR point cloud. BEV features are extracted via a ResNet34 backbone, and training is performed on the Navtrain split using 8 NVIDIA RTX 4090 GPUs. To enable controller-aware trajectory prediction, we build a controller-style reference bank capturing variations in controller parameters and post-dynamics perturbations, providing the Controller Style Extractor (CSE) with diverse examples for learning robust, controller-conditioned latent representations. Details of the perturbations are provided in the Appendix.

4.3 Main Results

NAVSIM. We evaluate WorldCAP on the NAVSIM [7] test set and compare it with state-of-the-art approaches, as summarized in Tab. 1. WorldCAP demonstrates consistent improvements across key driving metrics on NAVSIM [7]. Notably, it achieves higher performance in No At-Fault Collisions (NC, 98.7) and Drivable Area Compliance (DAC, 96.9), reflecting its ability to generate trajectories that more accurately respect drivable boundaries. By conditioning latent BEV state evolution on controller-specific execution characteristics via the Controller Style Encoder and CALA mechanism, WorldCAP captures systematic execution biases, actuator delays, and feedback-dependent vehicle responses that are typically neglected by conventional world models. This enables the system to anticipate discrepancies between nominal planned trajectories and their likely realized execution, producing more realistic latent state transitions. Consequently, downstream reward models receive more faithful trajectory evaluations, facilitating trajectory selection that aligns closely with real-world vehicle behavior and controller-specific dynamics.

Improvements in Ego Progress (EP, 82.3) and Time-to-Collision (TTC, 95.7) further demonstrate the benefits of execution-aware latent modeling. By accounting for how different controllers adjust acceleration and steering in response to deviations from planned trajectories, WorldCAP predicts both spatial and temporal evolution of vehicle states with higher fidelity. This allows the trajectory evaluation module to select paths that better preserve motion continuity, maintain realistic vehicle progression, and mitigate abrupt accelerations or decelerations, thereby enhancing TTC. Overall, this integrated execution-aware modeling produces latent state evolutions that closely reflect realistic vehicle responses, ensuring that the selected trajectories are consistent with controller-specific execution characteristics and represent a more faithful mapping from planned paths to executed trajectories.

4.4 Ablation Studies

Temporal Controller-Style Injection Depth. We investigate how the temporal injection of controller style influences future trajectory prediction in the

Table 1: Comparison of state-of-the-art methods on the NAVSIM test split. NC: No At-Fault Collisions; DAC: Drivable Area Compliance; EP: Ego Progress; TTC: Time-to-Collision; Comf.: Comfort; PDMS: Predictive Driver Model Score; Traj.: Trajectory; Eval.: Evaluation; C: Camera; L: LiDAR.

Method	Input	#Traj.	Traj. Eval.	NC↑	DAC↑	EP↑	TTC↑	Comf.↑	PDMS↑
Human	-	-	-	100	100	87.5	100	99.9	94.8
Constant Velocity	-	1	×	69.9	58.8	49.3	49.3	100.0	21.6
Ego Status MLP	-	1	×	93.0	77.3	62.8	83.6	100.0	65.6
VADv2	C	8192	Rule-based	97.9	91.7	77.6	92.9	100.0	83.0
UniAD	C	1	Rule-based	97.8	91.9	78.8	92.9	100.0	83.4
LTF	C	1	×	97.4	92.8	79.0	92.4	100.0	83.8
PARA-Drive	C	1	Rule-based	97.9	92.4	79.3	93.0	99.8	84.0
TransFuser	C & L	1	×	97.7	92.8	79.2	92.8	100.0	84.0
LAW	C	1	×	96.4	95.4	81.7	88.7	99.9	84.6
DRAMA	C & L	1	×	98.0	93.1	80.1	94.8	100.0	85.5
Hydra-MDP	C & L	8192	Model-free	98.3	96.0	78.7	94.6	100.0	86.5
WoTE	C & L	256	Model-based	98.5	96.8	81.7	94.9	100.0	88.2
ResWorld	C & L	256	Model-based	98.2	96.4	**82.5**	**98.9**	100.0	88.3
WorldCAP	C & L	256	Model-based	**98.7**	**96.9**	82.3	95.7	**100.0**	**89.1**

BEV latent world model. Specifically, we compare two strategies: injecting the controller style only at the initial transition step (step 0) followed by controller-agnostic rollout for the remaining 4 s (shallow injection), and injecting the controller style at every prediction step throughout the horizon (persistent injection). This experiment isolates whether a single conditioning signal is sufficient to encode controller-dependent dynamics or whether continuous conditioning is required during long-horizon prediction.

As shown in Tab. 2, persistent injection consistently improves trajectory quality and achieves a higher PDMS. When the controller style is injected only once, the conditioning signal influences only the first transition, while subsequent latent dynamics gradually revert to controller-agnostic behavior. This mismatch leads to inconsistent latent evolution during long-horizon rollout, resulting in degraded trajectory evaluation performance. Persistent temporal injection, in contrast, continuously conditions each latent transition on controller-specific characteristics, allowing the world model to maintain coherent controller-aware dynamics throughout the rollout. This enables the model to capture cumulative execution effects such as steering corrections and actuator delays, resulting in more consistent latent transitions and improved trajectory evaluation.

Inference Reference Count. We evaluate the robustness of the Controller Style Extractor (CSE) under different numbers of reference trajectories during inference, while keeping the training configuration fixed at 256 references. Specifically, we test 64, 128, and 256 reference trajectories to analyze how the size of the reference set affects trajectory evaluation.

As shown in Tab. 3, the model maintains strong performance across all settings, indicating that the learned controller-style representation generalizes well

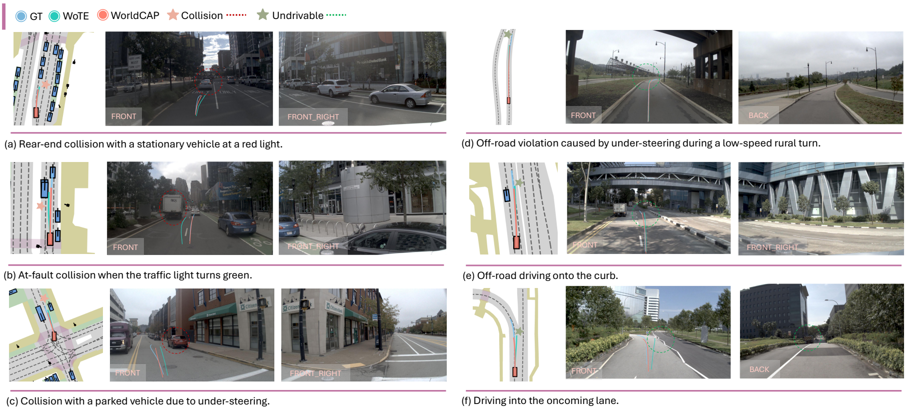

Fig. 3: Visualization of End-to-End Planning Results. We compare our method with WoTE [22], which serves as our trajectory planning baseline without controller style injection. (a)(b) In traffic light scenarios, WorldCAP effectively avoids stationary vehicles, whereas baseline methods exhibit minor rear-end collisions. (c)(d)(e) World-CAP keeps trajectories within drivable regions, reducing off-road violations or collison caused by under-steering. (f) In intersections, WorldCAP maintains safe separation and avoids collisions.

Table 2: Ablation on temporal controller-style injection. Persistent injection across all future steps produces more consistent trajectory evaluation and higher PDMS compared with single-step injection.

Injection Depth	NC↑	DAC↑	DDC↑	EP↑	TTC↑	Comf.↑	PDMS↑
Step 0 Only (Shallow)	98.2	96.5	100.0	80.6	94.3	100.0	87.4
Every Step (Persistent)	98.7	96.9	100.0	82.3	95.7	100.0	89.1

to varying reference counts. Even with only 64 reference trajectories, the performance degradation is minimal, suggesting that the CSE can reliably infer controller characteristics from a relatively small sample set.Increasing the number of reference trajectories further improves performance. Providing more references offers additional observations of the specific controller execution pattern, enabling the CSE to estimate the underlying style representation more stably and with lower estimation variance. As a result, the extracted controller embedding becomes more reliable, allowing the world model to generate more consistent controller-conditioned latent transitions. This leads to more accurate trajectory evaluation and gradually improves EP, TTC, and the overall PDMS.

Training Reference Count. We further analyze the effect of the number of reference trajectories used during training for the CSE. Increasing the number of references provides more executed trajectories generated under the same con-

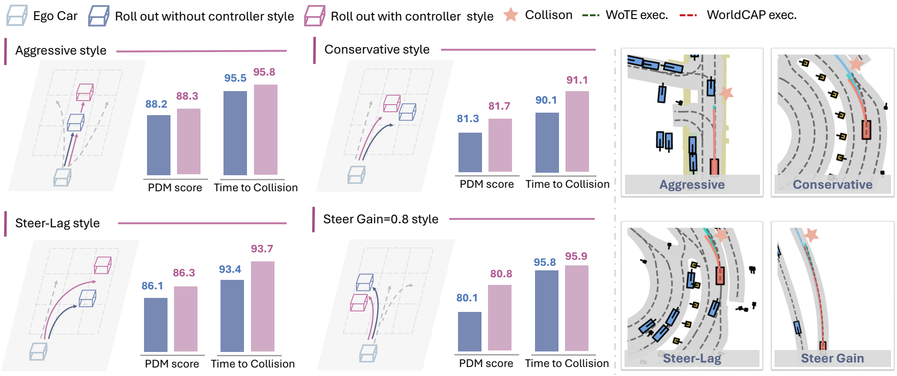

Fig. 4: Planning Results under Different Controller Styles. We evaluate WorldCAP across four representative controller styles, comparing its performance with WoTE [22]. The left panel shows improvements in PDMS, while the right panel presents representative trajectories. WorldCAP consistently adapts to diverse styles: (*Aggressive*) avoids overshooting; (*Conservative*) mitigates understeering; (*Steer-lag*) compensates for delayed steering; (*Reduced steering gain*) maintains lane adherence. Style-specific parameters are provided in the appendix. **WoTE exec.** denotes the trajectory executed by the controller when following the plan generated by WoTE, while **WorldCAP exec.** denotes the trajectory executed under the same controller when following the plan generated by WorldCAP.

troller style, supplying richer evidence for estimating the corresponding style representation. We evaluate three settings with 64, 128, and 256 reference trajectories, using the same number during both training and inference.

As shown in Tab. 4, performance improves consistently with more reference trajectories. With fewer references, the controller style is inferred from limited execution samples, resulting in higher estimation variance and less stable embeddings, which reduces the reliability of the latent world model's conditioning. In contrast, a larger set of reference trajectories provides more consistent evidence of the controller's characteristic behaviors, yielding a more accurate and stable style representation. This enables the world model to better capture controller-dependent execution dynamics and generate more faithful latent state transitions.

4.5 Visual Analysis

End-to-end planning trajectories. To highlight the improvements in end-to-end driving achieved through controller style injection, we compare the qualitative results of WorldCAP with WoTE [22] in Fig. 3. It is evident that the trajectories predicted by our method more effectively avoid collisions with other vehicles and curbs, particularly in scenarios involving turns or narrow roads. The

Table 3: Generalization of reference trajectories Count at Inference.The number of reference trajectories is the same during training but different during testing. Ref Traj.:Reference Trajectories.

# Ref Traj.	NC↑	TTC↑	EP↑	PDMS↑
64	98.6	95.2	81.5	88.3
128	98.5	95.2	82.0	88.6
256	98.7	95.7	82.3	89.1

Table 4: Ablation study on number of reference rrajectories During Training. The number of reference trajectories is the same during training and testing. Ref Traj.:Reference Trajectories.

# Ref Traj.	NC↑	TTC↑	EP↑	PDMS↑
64	97.6	92.7	80.3	87.0
128	98.6	95.1	81.5	88.3
256	98.7	95.7	82.3	89.1

predicted trajectories also remain within the feasible drivable region, adhering better to road boundaries and lane structures.

Different controller style. To evaluate robustness under heterogeneous execution behaviors, we analyze WorldCAP trajectories under four controller styles: *Aggressive*, *Conservative*, *Steer-lag*, and *Reduced steering gain (0.8)*. The first two correspond to tracker-level perturbations, while the latter two correspond to post-dynamic-level perturbations; detailed parameters are provided in the appendix. Each style induces distinct deviations in vehicle response, including delayed steering, high-frequency oscillations, over- or under-steering, and cautious maneuvers, which are effectively handled by our method. As shown in Fig. 4, WorldCAP generalizes well across these styles, consistently improving PDM scores. Specifically, the left panel presents the score improvements for each style as a bar chart, while the right panel visualizes representative trajectory examples under each controller style.

5 Conclusion

We introduce WorldCAP, a controller-adaptive framework for end-to-end autonomous driving that explicitly incorporates execution-dependent dynamics into latent BEV world modeling. By conditioning latent transitions on controller-specific execution characteristics, WorldCAP captures realistic deviations caused by heterogeneity of controllers, enabling execution-aware trajectory prediction without modifying the underlying planner or reward formulation.

Extensive evaluation on the NAVSIM benchmark demonstrates that WorldCAP consistently enhances trajectory fidelity, motion smoothness, and operational consistency across heterogeneous controllers. Complementary ablation studies highlight the critical roles of temporal conditioning depth and reference trajectory richness in enabling robust extraction of controller styles.

Overall, WorldCAP establishes a principled paradigm for bridging the gap between nominal planning and real-world execution, providing execution-aware trajectory predictions that advance both the reliability and generalization of end-to-end autonomous driving systems.

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